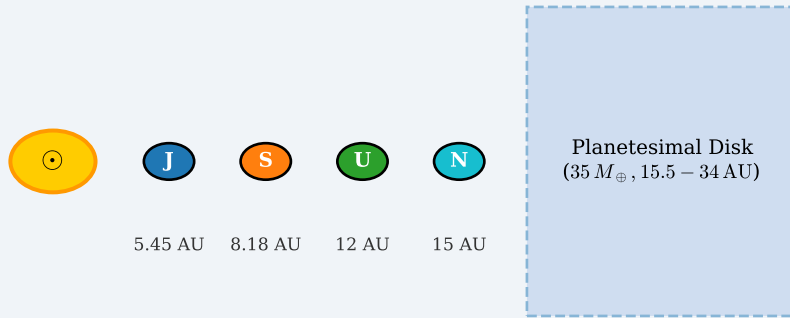
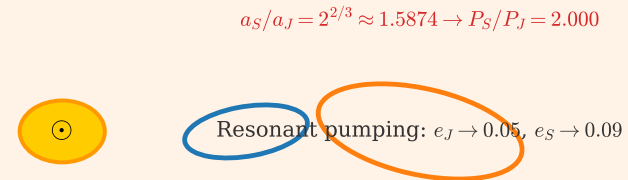


1. Primordial Compact Architecture ($t < 600 - 880$ Myr)



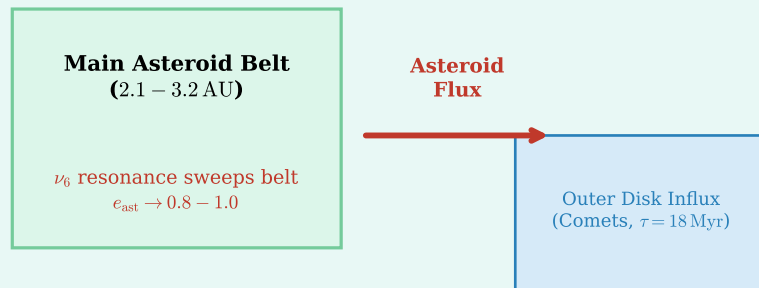
Slow planetesimal scattering $\rightarrow$ Jupiter in, Saturn out ($P_S/P_J: 1.83 \rightarrow 2.0$)

2. 2:1 MMR Crossing & Eccentricity Jump ($t = t_{\text{inst}}$)



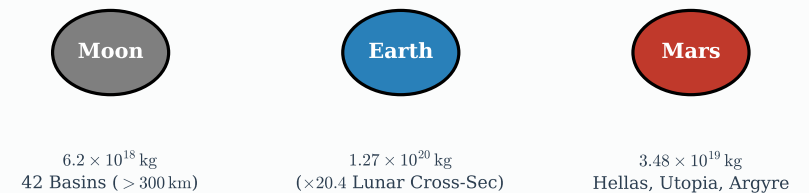
Destabilizes Ice Giants $\rightarrow$ chaotic encounters plunge Ice Giants into disk

3. Secular Resonance Sweep (ν_6, ν_5, ν_{16})



95% of asteroids destabilized $\rightarrow$ inner solar system impact corridor

4. Terrestrial Bombardment & Lunar Basins



Duration $\approx 100 - 150$ Myr with sharp peak matching Apollo $^{40}\text{Ar} - ^{39}\text{Ar}$ ages (3.85 – 3.95 Ga)